

2. Row 1 Leading Entry = 1 in Column 1
 Row 2 Leading Entry = 1 in Column 2
 $\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions below the leading entry must contain 0 is True.

4. The condition that all leading entries must be 1 is True.

5. Row 1 Column 2 = 0
 $\therefore$ The condition that for each column having a leading entry, all positions above the leading entry must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 9 & -8 \\ 0 & 1 & -4 & 5 \end{bmatrix}$ is in rref $\square$

By defn, there are no pivots in Column 3 and 4.

$\therefore$ By defn, x_3 and x_4 are free variables.

The corresponding system of equations are: ① $x_1 + 9x_3 - 8x_4 = 0$
 ② $x_2 - 4x_3 + 5x_4 = 0$
 $\therefore$ The solution set is: $x_1 = -9x_3 + 8x_4$, $x_2 = 4x_3 - 5x_4$

x_3 , x_3 and x_4 are free variables.

$\therefore$ The solution set in parametric vector form is: $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

$$= \begin{bmatrix} -9 \\ 4 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 8 \\ -5 \\ 0 \\ 1 \end{bmatrix} x_4 \quad \square$$

$$9. \begin{bmatrix} 3 & -9 & 6 \\ -1 & 3 & -2 \end{bmatrix} \xrightarrow[\times 1/3]{\text{Row 2} = \text{Row 1}} \begin{bmatrix} 1 & -3 & 2 \\ -1 & 3 & -2 \end{bmatrix} \xrightarrow[\text{+Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & -3 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1 is non-zero, Row 2 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero row is True.